DASHBOARD / KNOWLEDGE BASE

RA Family: How to Change the MTB (Micro Trace Buffer) settings with J-Link

Last Updated: 03/28/2025

Description

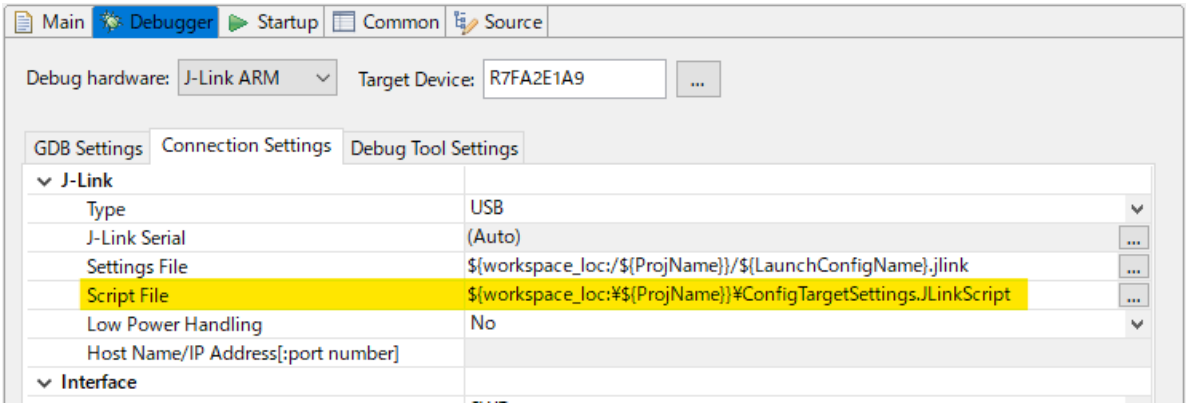
For MCUs with ARM® CoreSight ™ MTB (Micro Trace buffer) such as RA2E1, the MTB buffer may be mapped to the beginning of the SRAM area. When a user application uses the same SRAM area, it doesn't work. To avoid this issue, the memory allocation of the MTB buffer can be changed as follows.

1. Create a JLinkScript file with the following function. The file name extension should be ***JLinkScript***, including upper and lower case. In the following example, the MTB buffer is defined as the 1KB area from 0x20007C00.

```
int ConfigTargetSettings(void)
{
  JLINK_ExecCommand("CORESIGHT_SetMTBBufUseAddr=0x20007C00");
  JLINK_ExecCommand("CORESIGHT_SetMTBBufUsageSize=0x400");
  JLINK_ExecCommand("SelectTraceSource=2");
  return 0;
}
```

Note: The above version works for e2studio 2025-01 and 2024-10. Remove the SelectTraceSource line for older versions.

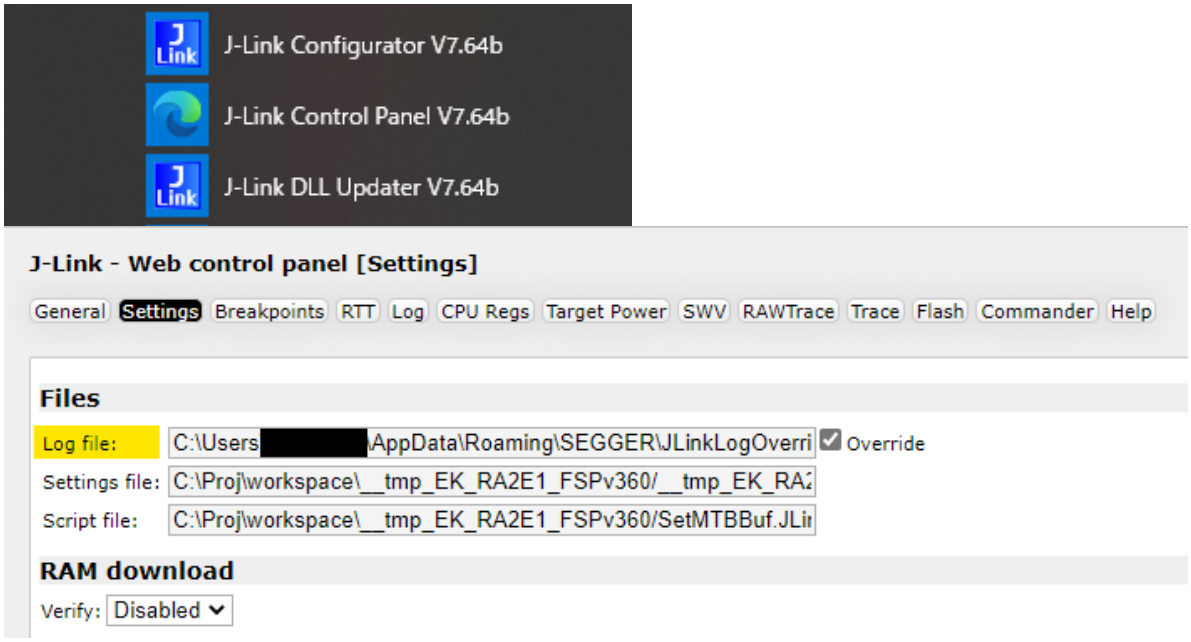
2. Set the JlinkScript file as the J-Link Script File.



3. Connect the target board with the J-Link. The MTB buffer setting can be confirmed in the ***JLinkLogOverride.log*** as follows.

```
T4F68 003:353.246   MTB: Starting trace using 0x400 bytes buffer @ 0x20007C00
```

The file location of *JLinkLogOverride.log* can be confirmed on the J-Link Control Panel. For Windows machines, select the J-Link Control Panel from the Start Menu, and find the “Log file:” in the Settings menu.



Suitable Products

FSP, J-Link
e2studio 2025-01 and 2024-10 (and older)
RA

Source:

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